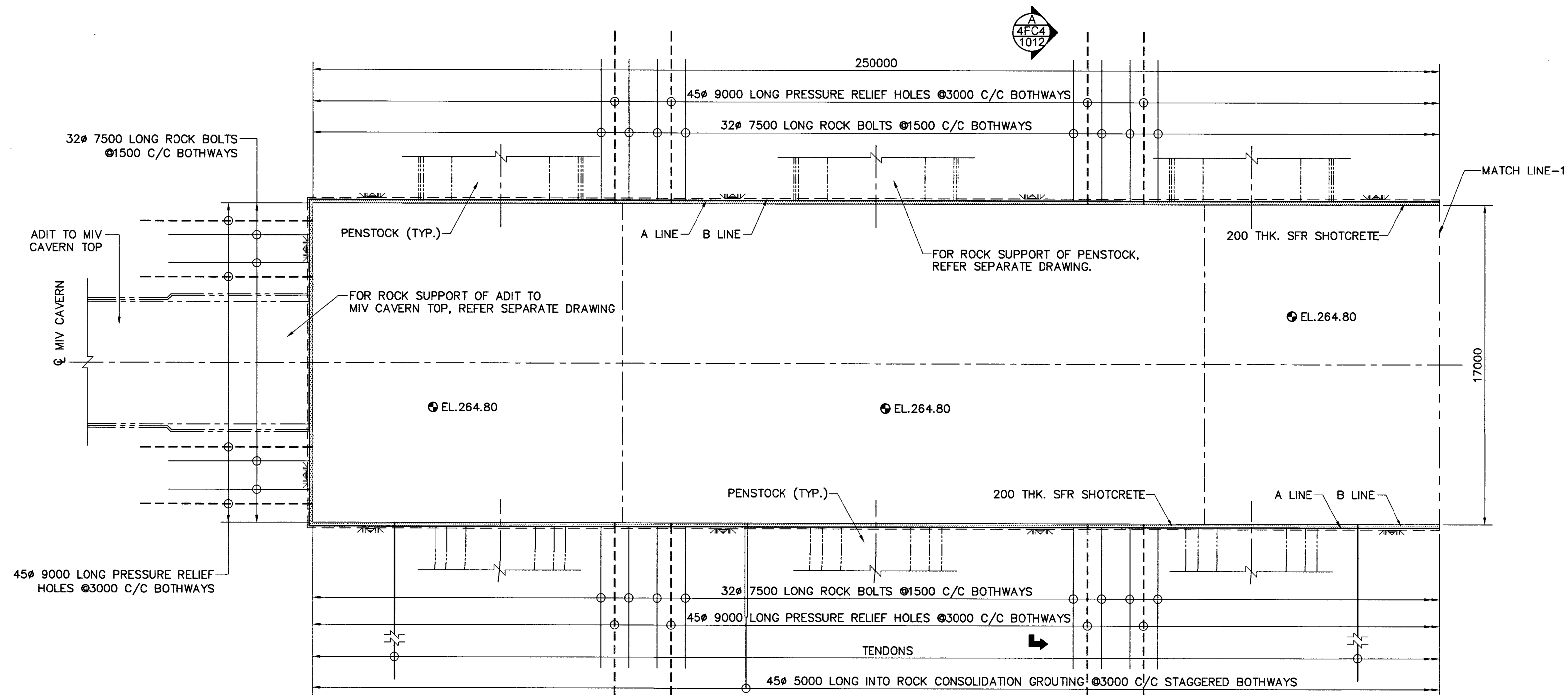
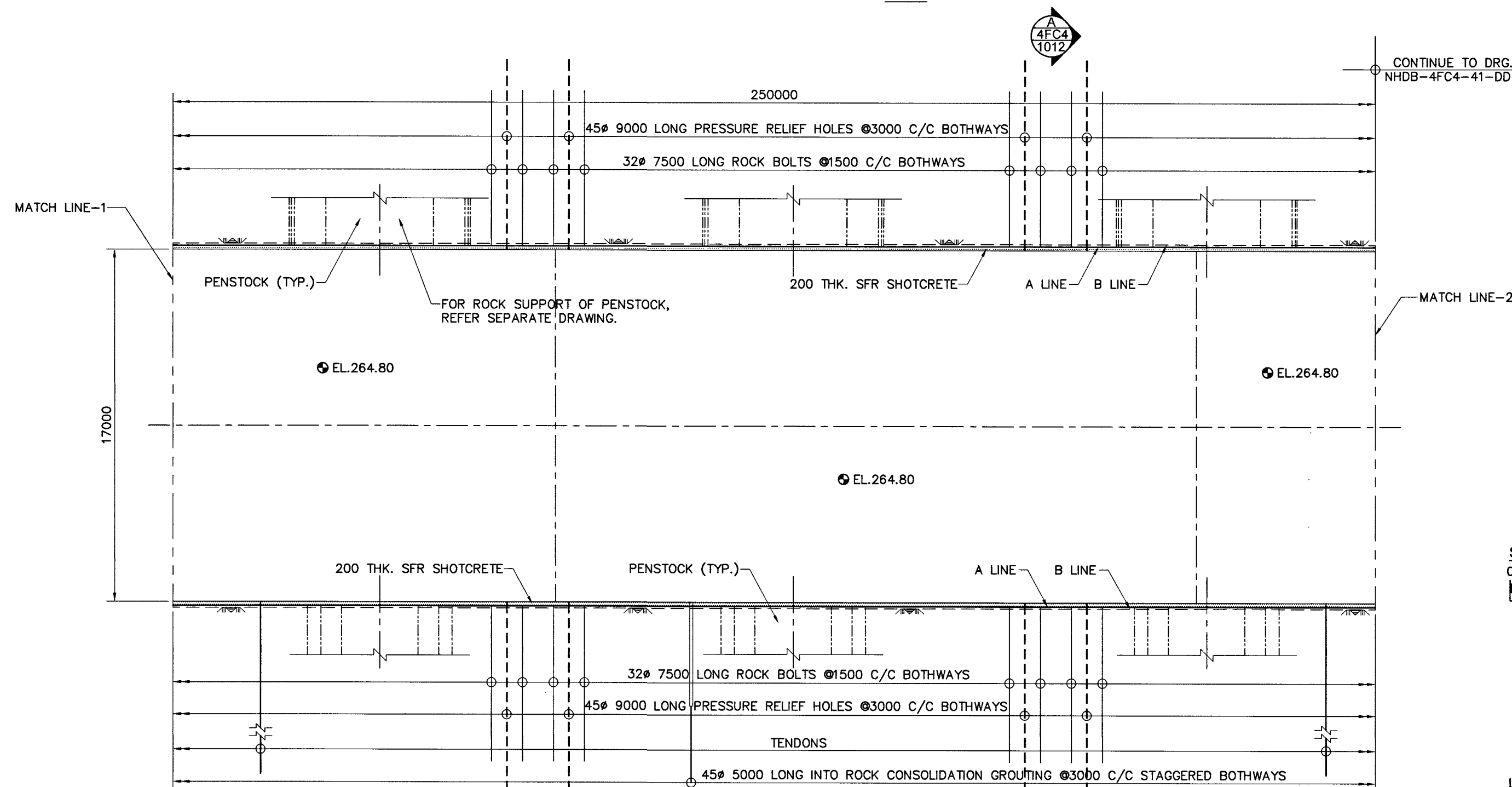




PLAN



PLAN

SCALE:

0 2.0 4.0 6.0 8.0 10.0 METRE

REFERENCE DRAWINGS:-

1. NHDB-4AC4-41-DD-0001 & 1003.
2. NHDB-4FC4-41-DD-1001.
3. NHDB-4FC4-41-DD-1002.
4. NHDB-4FC4-41-DD-1006 TO 1010.

31.03.2026	00	ISSUED FOR CONSTRUCTION	BY	CH/APP	DATE	MAR 2026	DRG. NO.	NHDB	4FC4	41	DD	1011	00
DATE	NO.	REVISION OR ISSUES	BY	CH/APP	DATE	MAR 2026	DRG. NO.	NHDB	4FC4	41	DD	1011	00
DESIGN MEMO NO.													
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NOTES:

1. सभी माप मिलीमीटर में तथा ऊंचाई मीटर में दिए गए हैं।
2. THIS DRAWING IS ISSUED FOR EXCAVATION & ROCK SUPPORT OF MIV CAVERN EXCAVATION FROM EL.286.00 TO EL.264.80. FOR LOCAL EXCAVATION & ROCK SUPPORT BELOW EL.264.80, REFER SEPARATE DRAWING.
3. EXCAVATION SHALL BE DONE IN STAGES CONFORMING TO SECTION B.3.8.7 OF TECHNICAL SPECIFICATIONS. BEFORE TAKING UP THE EXCAVATION OF CAVERN, EXCAVATION METHODOLOGY SHALL BE SUBMITTED BY THE CONTRACTOR AND APPROVED BY THE ENGINEER-IN-CHARGE. THE EXCAVATION SEQUENCE SHALL BE GUIDED AS PER ISSUED DRAWING NO. NHDB-4AC4-41-DD-1003.
4. EXCAVATION SEQUENCE MAY MODIFY TO SUIT THE SITE CONDITIONS.
5. THE DEFINITION OF 'A' AND 'B' LINE SHALL BE AS PER SECTION B.3 OF TECHNICAL SPECIFICATIONS.
6. NEXT STAGE EXCAVATION SHALL START ONLY AFTER INSTALLATION OF ROCK SUPPORT & INSTRUMENTS IN THE PREVIOUS STAGE INCLUDING GROUTING OF THE ROCK BOLTS.
7. THIS DRAWING SHOWS FINAL ROCK SUPPORT SYSTEM ONLY. IN ADDITION SOME TEMPORARY SUPPORTS MAY BE REQUIRED DURING EXCAVATION BEFORE INSTALLATION OF THE FINAL SUPPORTS. THE SAME MAY BE INSTALLED DEPENDING UPON SITE CONDITIONS WITH THE APPROVAL OF ENGINEER-IN-CHARGE.
8. IN MIV CAVERN (D/S WALL) BETWEEN EL.288.25 TO EL.264.80, 45mm 5000 LONG (INTO ROCK) CONSOLIDATION GROUTING @3000 C/C STAGGERED BOTHWAYS SHALL BE PROVIDED THROUGHOUT AS PER SECTION B.7 OF TECHNICAL SPECIFICATIONS. IN MIV CAVERN (U/S WALL) BETWEEN EL.288.25 TO EL.264.80 45mm 5000 LONG (INTO ROCK) CONSOLIDATION GROUTING @3000 C/C STAGGERED BOTHWAYS SHALL BE PROVIDED IN WEAK ROCK MASS AS DIRECTED BY ENGINEER-IN-CHARGE, AS PER SECTION B.7 OF TECHNICAL SPECIFICATIONS.
9. THE LOCATION, ORIENTATION AND DEPTH OF THE DRILL HOLES FOR THE CONSOLIDATION GROUTING ARE TENTATIVELY INDICATED ON THE DRAWINGS. THE FINAL NO. LENGTH, LOCATION AND INCLINATION OF THE DRILL HOLES AS WELL AS THE COMPOSITION AND CONSISTENCY OF THE GROUT MIXES, GROUTING PRESSURE, PUMPING RATES AND SEQUENCE IN WHICH THE HOLES ARE TO BE DRILLED AND GROUTED SHALL BE GOVERNED BY ACTUAL CONDITION ENCOUNTERED ON SITE AND SHALL BE AT ALL TIMES SUBJECT TO THE APPROVAL BY THE ENGINEER-IN-CHARGE.
10. LINE DRILLING SHALL BE USED ALONG WITH CONTROL BLASTING FOR EXCAVATION TO ACHIEVE THE DESIRED EXCAVATION PROFILE. LINE DRILLING SHALL BE CARRIED OUT CONFORMING TO CLAUSE 3.8.2.2 OF SECTION B.3 OF TECHNICAL SPECIFICATION AS DIRECTED BY ENGINEER-IN-CHARGE.
11. THE BENCHING EXCAVATION SHALL IN NO CASE BE MORE THAN 3000 HEIGHT UNLESS IT IS SUPPORTED BY THE PERMANENT SUPPORT SHOWN IN THE DRAWINGS.
12. THE ROCK BOLTS SHOWN IN THE DRAWING SHALL BE CONFORMING TO SECTION B.4 OF TECHNICAL SPECIFICATIONS. THE ROCK BOLTS SHALL BE TENSIONED AS PER CLAUSE 4.5.8 OF SECTION B.4 OF TECHNICAL SPECIFICATIONS. MINOR MODIFICATION IN THE LENGTH, ORIENTATION, PATTERN AND LOCATION OF ROCK BOLTS MAY BE DONE TO SUIT THE SITE CONDITIONS WITH THE APPROVAL OF ENGINEER-IN-CHARGE. ADDITIONALLY, IN CASE OF PROMINENT SHEAR ZONES, IF ANY, PRE GROUTING SHALL BE PERFORMED AS PER SITE CONDITION IN CONSULTATION WITH SITE GEOLOGIST AS PER B.4 OF TECHNICAL SPECIFICATION AND AS APPROVED BY ENGINEER-IN-CHARGE.
13. IN CASE OF WEAK ROCK MASS WHERE END ANCHORAGE WITH EXPANSION SHELL ROCK BOLT IS NOT POSSIBLE, RESIN GROUTED ROCK BOLTS SHALL BE PROVIDED IN CONSULTATION WITH SITE GEOLOGIST AS PER DIRECTION OF ENGINEER-IN-CHARGE.
14. THE ROCK BOLTS SHALL BE INSTALLED ONLY ON FINISHED EXCAVATION SURFACES UNLESS AS GOVERNED BY GEOLOGICAL CONDITION AS APPROVED BY ENGINEER-IN-CHARGE.
15. THE CONTRACTOR SHALL EXERCISE ALL PRECAUTIONS TO PREVENT DAMAGE TO ALREADY PLACED SHOTCRETE DURING THE BLASTING FOR THE NEXT PHASE OF EXCAVATION.
16. ATLEAST 50% BOLTS SHALL FORM THE PART OF CYCLE. BALANCE BOLTS CAN BE INSTALLED OUT OF CYCLE BUT BEFORE COMPLETING A PARTICULAR STAGE OF EXCAVATION. THE SPACING BETWEEN THE TWO BOLTS SHALL IN NO CASE EXCEED 3000.
17. INSTALLATION METHODS OF ROCK BOLTS AND SHOTCRETE SHALL BE FULLY OPTIMIZED THROUGH TEST BOLTS IN THE REPRESENTATIVE ROCK, PRIOR TO ACTUAL START OF WORK.
18. SFR SHOTCRETE SHOWN IN THE DRAWING SHALL BE APPLIED AS SPECIFIED IN SECTION B.5 OF TECHNICAL SPECIFICATION. IF REQUIRED, PLAIN SHOTCRETE WITH WIREMESH CAN BE USED AS PER DIRECTION OF ENGINEER-IN-CHARGE.
19. THE PRESSURE RELIEF HOLES SHOWN IN THE DRAWING SHALL CONFORM TO B.7 OF TECHNICAL SPECIFICATIONS. 36mm SLOTTED PVC LINERS IN PRESSURE RELIEF HOLES SHALL BE INSTALLED AS PER THE DIRECTION OF ENGINEER-IN-CHARGE.
20. FOR DETAILS OF PRESSURE RELIEF HOLES ARRANGEMENTS AND INSTRUMENTATION REFER DRAWING NO. NHDB-4FC4-41-DD-1006 TO 1010.
21. POST TENSIONED ROCK BOLTS (TENDONS), THEIR INSTALLATION, TENSIONING AND TESTING SHALL CONFIRM TO SECTION B.4 OF TECHNICAL SPECIFICATION. FOR TYPE, LOCATION AND LEVELS OF TENDONS REFER DRAWING NO. NHDB-4FC4-41-DD-1006 TO 1010.

एन एच पी सी लिमिटेड (भारत सरकार का उद्यम) NHPC Limited (A GOVT. OF INDIA ENTERPRISE)	
दिवान बहुउद्देशीय परियोजना DIBANG MULTIPURPOSE PROJECT	
MIV CAVERN EXCAVATION AND ROCK SUPPORTS FROM EL. 286.00M TO EL. 264.80M	
SHEET 1 OF 2	
31.03.2026	00
DATE	NO.
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DD	1011
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